

Name:	B/O Sasi Priya	Case ID:	50800110018
Age:	3 days	Sample Type:	Peripheral Venous Blood
Sex:	Female	Sample Collection Date:	22 Aug 2025
Referring Clinician:	Dr. S Kiruthika	Sample Receipt Date:	23 Aug 2025
Test Required:	Exome First	Report Date:	13 Sep 2025
Location:	Salem Genetics Center,Salem	Version:	I

CLINICAL INFORMATION/HISTORY

SASI PRIYA is non-consanguineously married to MUNI PRASAD. The couple's first pregnancy was a missed abortion. G2- female baby died at 3 months (sudden death) and was suspected to have increased startle response, seizures, and cardiomegaly. G3- female baby, intra uterine death at 32 weeks of gestation, maternal history of chicken pox at 23-24 weeks of gestation. G4- missed abortion. G5- female baby born through normal vaginal delivery, no history of NICU stay. Neurosonogram of the baby was Normal, USG abdomen- normal. Normal limbs, suspected prominence of clitoris. Couple karyotyping report normal. The baby (G5) is being evaluated for gene variations relating to the reported phenotype.

The following HPO terms were used for analysis: inborn errors of metabolism, sudden death, neonatal death, Exaggerated startle response, seizures, cardiomegaly, intra uterine death, abnormal size of the clitoris, abnormal clitoris morphology, Abnormal size of the clitoris

RESULT SUMMARY

A heterozygous likely pathogenic variant for a recessive disorder causative of the reported phenotype was identified

*Correlation with clinical profile and family history is required.

Summary of Findings

Variants Potentially Relevant to the Indication for Testing:

The index patient is:

- Heterozygous for a Likely Pathogenic variant in the *DMGDH* gene associated with DIMETHYLGLYCINE DEHYDROGENASE DEFICIENCY; DMGDHD .
- Heterozygous for a Likely Pathogenic variant in the *GATM* gene associated with CEREBRAL CREATINE DEFICIENCY SYNDROME 3; CCDS3 .
- Heterozygous for a Likely Pathogenic variant in the *PPA2* gene associated with SUDDEN CARDIAC FAILURE, INFANTILE; SCFI SUDDEN CARDIAC FAILURE, ALCOHOL-INDUCED; SCFAI .

Carrier Status:

- The index patient is a carrier of a Pathogenic variant in the *GALT* gene associated with GALACTOSEMIA I; GALAC1 .

Secondary Findings (ACMG gene list):

- No Pathogenic or Likely Pathogenic (Class 1/2) variants were detected in the ACMG gene list.

FINDINGS RELATED TO PHENOTYPE

Gene&Transcript	Variant	Location	Zygosity	Disorder(OMIM)	Inheritance	Variant Classification
DMGDH NM_013391.3	c.102-2A>G	Intron 1	Heterozygous	DIMETHYLGLYCINE DEHYDROGENASE DEFICIENCY; DMGDHD:605850	Autosomal Recessive	Likely Pathogenic
GATM NM_001482.3	c.242dup p.Arg82GlnfsTer20	Exon 2	Heterozygous	CEREBRAL CREATINE DEFICIENCY SYNDROME 3; CCDS3:612718	Autosomal Recessive	Likely Pathogenic
PPA2 NM_176869.3	c.500C>T p.Pro167Leu	Exon 6	Heterozygous	SUDDEN CARDIAC FAILURE, INFANTILE; SCFI:617222	Autosomal Recessive	Likely Pathogenic

* Genomic Position based on Assembly GRCh37, **Number of applied in silico programs predicting the effect of the variant on the protein outcome (CADD: Combined Annotation Dependent Depletion (v1.6), SIFT, PolyPhen-2,

MT: Mutation Taster), N/A: Not Applicable, ***Minor Allele Frequency as described in GnomAD (Controls), ****based on ACMG Guidelines. het=heterozygous, hom=homozygous, hemi=hemizygous.

VARIANT INTERPRETATION

DMGDH chr5:78359612 (T/C) - Likely Pathogenic

A Heterozygous 3' splice site variation in intron 1 of the *DMGDH* gene (c.102-2A>G) that affects the invariant AG acceptor splice-site of exon 2 was detected. The observed variant has a minor allele frequency of 0.00001 in gnomAD exomes and novel in genomes, respectively. The splice AI score of the variant is 0.63. **Based on the above evidence, this variant has been classified as Likely Pathogenic according to the ACMG guidelines.**

However we did not identify the second significant heterozygous variations in *DMGDH* gene and there is no significant heterozygous deletion and/or duplication detected in the gene by Copy Number Variation (CNV) algorithm

Dimethylglycine dehydrogenase deficiency (DMGDHD) is an inborn error of metabolism characterized by a fish-like odor, chronic fatigue, and increased level of the muscle form of creatine kinase in serum.

GATM chr15:45668844 (G/GC) - Likely Pathogenic

A Heterozygous Frameshift variant c.242dup in Exon 2 of the *GATM* gene that results in a frameshift and premature truncation of the protein downstream to codon 82 (p.Arg82GlnfsTer20) was identified. The observed variant is novel in gnomAD exomes. The variant is predicted to lead to nonsense-mediated decay of the transcript. **Based on the above evidence, this variant has been classified as Likely Pathogenic according to the ACMG guidelines.**

However we did not identify the second significant heterozygous variations in *GATM* gene and there is no significant heterozygous deletion and/or duplication detected in the gene by Copy Number Variation (CNV) algorithm

Cerebral creatine deficiency syndrome-3 (CCDS3) is an autosomal recessive disorder characterized by developmental delay/regression, mental retardation, severe disturbance of expressive and cognitive speech, and severe depletion of creatine/phosphocreatine in the brain. Most patients develop a myopathy characterized by muscle weakness and atrophy later in life. Oral creatine supplementation can offer symptom improvement.

PPA2 chr4:106359135 (G/A) - Likely Pathogenic

A Heterozygous Missense variant c.500C>T in Exon 6 of the *PPA2* gene that results in the amino acid substitution p.Pro167Leu was identified. The observed variant has a minor allele frequency of 0.00002 in gnomAD exomes and novel in genomes, respectively. The severity of the impact of this variant on the protein is high, based on the effect of the protein and REVEL (0.876), CADD-PHRED (31), ALPHA-MISSENSE (0.45630). ClinVar has also classified this variant as LikelyPathogenic[Variant ID:372227].The missense c.500C>T (p.Pro167Leu) variant in *PPA2* gene has been reported in individuals affected with infantile sudden cardiac failure (Kennedy H, et. al., 2016). **Based on the above evidence, this variant has been classified as Likely Pathogenic according to the ACMG guidelines.**

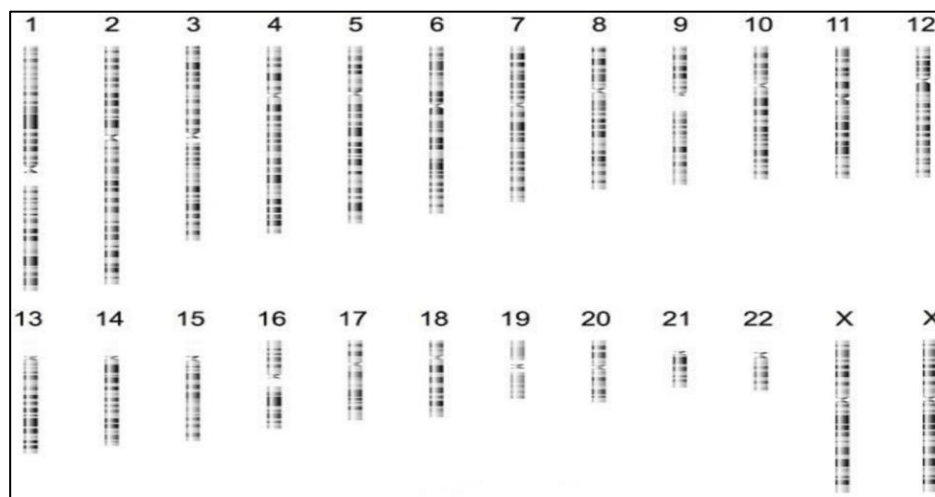
However we did not identify the second significant heterozygous variations in *PPA2* gene and there is no significant heterozygous deletion and/or duplication detected in the gene by Copy Number Variation (CNV) algorithm

Infantile sudden cardiac failure (SCFI) is caused by homozygous or compound heterozygous mutation in the *PPA2* gene on chromosome 4q24.Clinical presentation involved lactic acidosis, seizures, hypotonia, and cardiac dysrhythmia within the first months of life. Myocyte loss, disarray, or fibrosis was present in all individuals.

CNV/SV FINDINGS RELATED TO PHENOTYPE

Gene & Transcript	Variant	Disorder (OMIM)	Classification
No CNV/SV findings associated with the patient phenotype.			

KARYOVIEW



*Karyoview is a visual representation of chromosomal alterations observed in the sample. GAIN is represented in blue, LOSS is represented in red. Absence of these indicates a negative finding in the sample. Sex chromosomes will not be displayed for a prenatal sample (except for sex chromosomal abnormalities).

RECOMMENDATIONS

- The interpretation of this result should be done in the context of this individual's clinical and biochemical profile. Genetic counseling is recommended.

FINDINGS UNRELATED TO PHENOTYPE

Diagnostic findings not related to phenotype: No pathogenic variants in genes that are unrelated to the patient's phenotype were detected in this individual.

Carrier status in genes related to disease:

Gene&Transcript	Variant	Zygosity	Disorder(OMIM)	Inheritance	Variant Classification
GALT NM_000155.4	c.-129_-126del	Heterozygous	GALACTOSEMIA I; GALAC1:230400	Autosomal Recessive	Pathogenic

* Genomic Position based on Assembly GRCh37, **Number of applied in silico programs predicting the effect of the variant on the protein outcome (CADD: Combined Annotation Dependent Depletion (v1.6), SIFT, PolyPhen-2,

MT: Mutation Taster), N/A: Not Applicable, ***Minor Allele Frequency as described in GnomAD (Controls), ****based on ACMG Guidelines. het=heterozygous, hom=homozygous, hemi=hemizygous.

VARIANT INTERPRETATION

GALT chr9:34646572 (CCAGT/C) - Pathogenic

A Heterozygous Upstream variant c.-129_-126del of the *GALT* gene was identified. The observed variant has a minor allele frequency of 0.04889 in gnomAD exomes and novel in genomes, respectively. The severity of the impact of this variant on the protein is low, based on the effect of the protein. ClinVar has also classified this variant as Pathogenic (Variation ID: 25111). This variant has been observed in individuals affected with Galactosemia (Trbusek M, et al., 2001). Experimental study demonstrated that the c.-129_-126del (Alternative Position: c.-119_-116delGTCA) variant confers reduced promoter activity to transfected cell lines and also showed that human lymphoblasts derived from patients with the Duarte allele have reduced *GALT* mRNA, leading to reduced *GALT* activity (Elsas LJ, et al., 2001). **Based on the above evidence, this variant has been classified as Pathogenic according to the ACMG guidelines.**

Galactosemia I (GALAC1), or classic galactosemia, is an autosomal recessive disorder of galactose metabolism. Most patients present in the neonatal period, after ingestion of galactose, with jaundice, hepatosplenomegaly, hepatocellular insufficiency, food intolerance, hypoglycemia, renal tubular dysfunction, muscle hypotonia, sepsis, and cataract. Long-term complications include mental retardation, verbal dyspraxia, motor abnormalities, and hypergonadotropic hypogonadism.

Familial Risk:

The index patient is heterozygous for this variant. Carriers of this variant do not exhibit symptoms and have a 50% statistical risk of passing on the variant to any future offspring. Parents who are both carriers of this variant have a 25% (or 1 in 4) risk for having a child who is likely to be affected

SECONDARY FINDINGS

No pathogenic variants in genes related on the ACMG recommended secondary list were detected in this individual (Miller et al., Genet Med. 2021 May 20. PMID: 34012068)

Reported Variant Statistics

GeneTranscript	Variant	Location	Depth	Allelic Depth	Alternate Allele Fraction	dbSNP rsID
GALT NM_000155.4	c.-129_-126del	Upstream	53	24	54.72%	rs111033640
DMGDH NM_013391.3	c.102-2A>G	Intron 1	60	26	56.67%	rs764395701
GATM NM_001482.3	c.242dup p.Arg82Glnfs Ter20	Exon 2	70	40	42.86%	-
PPA2 NM_176869.3	c.500C>T p.Pro167Leu	Exon 6	60	34	43.33%	rs546693824

Data Quality Statistics:Total Data Generated (Gb): 8, Panel Coverage %: 99, Q30 bases: >85%

Mean Depth (X)	% Based Covered (1X)	(20X)	(30X)
85.89	98.37	97.26	96.14

Lifecell's Mendelian Variant classification (Based on ACMG recommendations)

- Class 1 - Pathogenic
- Class 2 – Likely Pathogenic
- Class 3 – Variant of uncertain significance
- Class 4 – Likely Benign
- Class 5 - Benign
- Class 6 – Disease associated SNP

The classification of variants is continually changing over time as more information becomes available. Please contact Lifecell in the future to determine if there have been any changes in classification of any reported variants.

METHODOLOGY

Sequencing of the protein-coding regions of approximately 30Mb of the human exome (targeting approximately 99% of regions in CCDS and RefSeq) and complete mitochondrial genome sequencing was performed using massively parallel next-generation sequencing (NGS) systems at a mean depth of 80-100X with the percentage of bases covered at 20X depth >90% in the target region and the depth for mitochondrial genome at 1000-2000X. In some cases, due to the complexity of the sequence, not all variants in the flanking regions are able to be analysed. A base is considered to have sufficient coverage at 20X and an exon is considered fully covered if all coding bases plus three nucleotides of flanking sequence on either side are covered at 20X or more. GATK best practice framework was followed for variant identification. Duplicate reads identification and removal, Base quality recalibration, and re-alignment of reads based on indels were done using the DRAGEN BIO IT platform (Version 4.2.4) (2). Quality checks (QC) will be performed on all VCF files to exclude variants where sequencing is of poor quality. Additional QC metrics include total homozygous and heterozygous calls (SNVs and indels), proportion of variant calls that were common, number of variants falling into different annotated consequence categories, and number of extreme heterozygotes (alternate allele proportion 0.8). Variant annotations will be done using published databases like OMIM, GWAS, GNOMAD, 1000Genome, etc (3-6). Non-synonymous and splice site variants will be used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported. Geneyx was used for variant annotation, analysis, and reporting (7).

LIMITATIONS

- The results specifically pertain to the examined sample and may not accurately represent the fetal chromosome constitution in instances of confined placental mosaicism or when the sample is contaminated with maternal cells.
- Despite taking all necessary precautions during DNA-based tests, the technical error rate for all types of DNA analysis is approximately 2%. Therefore, it is crucial to interpret all results within this context before taking any action based on these results.
- The report is generated within a specific timeframe known as the turnaround time (TAT) once the sample is received at the lab. However, the actual TAT may differ based on the complexity of the requested test(s) and information provided along with the sample. Lifecell cannot be held responsible for any delays that occur beyond the mentioned TAT.
- In certain rare cases, genetic tests may not provide accurate results, for example, when the quality of the sample given to Lifecell is not optimal. If a test performed by Lifecell fails due to unforeseen or unknown reasons beyond their control, Lifecell cannot be held responsible for any incomplete, potentially misleading, or incorrect results that were not foreseeable beforehand.
- Lifecell Pvt. Ltd has validated the test and determined its performance characteristics in accordance with the CAP/ACMG and NABL guidelines. All investigations have their limitations which are imposed by the limits of sensitivity & specificity of individual assay procedures as well as the quality of the specimen received by the laboratory.
- Clinical interpretation of given test result should be evaluated within the context of the patients medical history and other diagnostic laboratory test results.
- The present report comprises genetic analysis of the sample provided. It is important to note that this report cannot be accurately interpreted without information regarding clinical features and other laboratory reports, investigations. The information contained herein should not be considered a substitute for professional medical advice or diagnosis. It is highly recommended to consult with a qualified healthcare professional or medical specialist for a comprehensive evaluation and interpretation of your specific medical condition.

Version Set

Lifecell uses the most recent versions of the softwares, databases and Guidelines.

Version Set v6.2							
1kGenome	2013-05	ACMG	2025-06	AcmgSv	2025-06	AlphaMisse nse	v2-t114
CADD	v1.7	Camouflage dGenes	v1	Civic	2025-06-01	ClinGen	2024-12
ClinVar	2025-06-08	Converge	2017-04	Cytogenetic	2009-06	DarkGenes	v1
DbNsfP	4.4a	DbScSNV	1.1	DbSnp	b154	DGVGold	v107_2016- 05-15
DGVSV	v107_2020- 02-25	ESP6500	2	GeneEnhanc erSv	v6.2	GeneyxRepe ats	v1.2
Gerp	2010	gnomAD- exomes	2.1.1	gnomAD- genomes	2.1.1	gnomAD- mit	2.1.1
gnomAD- mnv	2.1.1	GnomADS V	v2.1	LitVar2	2025-05	MANE	v1.4
MasterMind	2025.04.19	MitoMap	2021-10	OMIM	2025-06	Phylop	2014-02
Revel	2016-03	Rmsk	2020-03	SnpEff	v5.2b.gx3- 2025-06	SpliceAI	1.3

DISCLAIMER

- Genetic testing is an important component of the diagnostic process, although it may not always yield definitive results. In certain situations, genetic tests fail to identify a specific genetic mutation/Variant, even if it is present. This can be attributed to the current constraints in medical understanding and testing capabilities. Accurate interpretation of test results may require knowledge about the true biological relationships in a family. Failing to accurately state the biological relationships in a family may result in incorrect interpretation of results, incorrect diagnoses, and/or inconclusive test results.
- NGS testing is not a substitute for professional medical advice or care. The results of NGS testing should always be interpreted in conjunction with a comprehensive medical evaluation and consultation with an expert healthcare professional.
- Genetic variations that have a potential link to the individual's medical condition, the interpretation of test results takes into consideration clinical findings, family history, and other laboratory data. It is important to note that rare polymorphisms sometimes result in false negative or positive results. Additionally, if the information provided is inaccurate or incomplete, there is a possibility of misinterpreting the results.
- Targeted clinical exome sequencing may not be able to detect specific events such as copy number variations, translocations, repeat expansions, and chromosomal rearrangements. Additionally, variants in untranslated regions, promoters, and intronic variants are not assessed using this method.
- The laboratory has interpreted the variants in the report to the best of their knowledge based on the information available at the time of reporting. However, the classification of variants may change over time, and Lifecell cannot be held responsible for any changes in classification.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- This test has certain limitations due to technology constraints, which means that not all parts of the exome can be fully analyzed. As a result, it is possible that certain variants in areas with insufficient coverage may not be identified or properly interpreted. This means that there may be pathogenic variants in the genes analyzed that were not detected by this test. These undetected variants could potentially affect the individual's phenotype, and it is recommended to confirm these findings using an alternative method.
- Though genetic testing was done, it is still possible that the person may have a genetic mutation in a gene that was not tested due to lack of information about their symptoms or medical history.
- Mutations have not been confirmed by Sanger sequencing.
- Genes with pseudogenes may have low sensitivity and specificity for variant detection and interpretation due to inability of data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- If Test results are alarming or unexpected, the client is advised to contact customer care immediately for possible remedial action.
- Test result released pertains to the specimen submitted. All test results are dependent upon the quality of the sample received by the Laboratory. Laboratory investigations are only tools to facilitate in arriving at a diagnosis and should be clinically correlated by the physician. Report delivery may be delayed due to unforeseen circumstances. Certain tests may require further tests at additional cost for derivation of exact value. Test reports may show interlaboratory variations. Contact Customer care for any report related concern within 24 hours of the receipt of the report.

REFERENCES

1. Richards S. et al., Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology, *Genet Med.*, 17(5):405-24, 2015.
2. https://en.mgi-tech.com/producats/instruments_info/5/
3. Landrum M. J. et al., ClinVar: public archive of interpretations of clinically relevant variants. *Nucleic Acids Res.*, 44(D1):D862-8, 2015.
4. Welter D. et al., The NHGRI GWAS Catalog, a curated resource of SNP-trait associations. *Nucleic Acids Res.*, 42:D1001-1006, 2014.
5. 1000 Genomes Project Consortium et al., A global reference for human genetic variation. *Nature*, 526(7571): 68-74, 2015.
6. <https://gnomad.broadinstitute.org/>
7. <https://analysis.geneyx.com/>
8. Carrier screening for genetic conditions. Committee Opinion No. 691. American College of Obstetricians and Gynecologists. *Obstet Gynecol* 2017;129:e41-55.
9. Miller DT, Lee K, Chung WK, Gordon AS, Herman GE, Klein TE, Stewart DR, Amendola LM, Adelman K, Bale SJ, Gollob MH, Harrison SM, Hershberger RE, McKelvey K, Richards CS, Vlangos CN, Watson MS, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.0 list for reporting of secondary findings in clinical exome and genome sequencing: a policy statement of the American College of Medical Genetics and Genomics (ACMG). *Genet Med.* 2021 May 20
10. Trbusek M, Francová H, Kozák L. Galactosemia: deletion in the 5' upstream region of the GALT gene reduces promoter efficiency. *Hum Genet.* 2001 Jul;109(1):117-20.
11. Elsas LJ, Lai K, Saunders CJ, Langley SD. Functional analysis of the human galactose-1-phosphate uridylyltransferase promoter in Duarte and LA variant galactosemia. *Mol Genet Metab.* 2001 Apr;72(4):297-305
12. Kennedy H, Haack TB, Hartill V, Mataković L, Baumgartner ER, Potter H, Mackay R, Alston CL, O'Sullivan S, McFarland R, Connolly G, Gannon C, King R, Mead S, Crozier I, Chan W, Florkowski CM, Sage M, Höfken T, Alhaddad B, Kremer LS, Kopajtich R, Feichtinger RG, Sperl W, Rodenburg RJ, Minet JC, Dobbie A, Strom TM, Meitinger T, George PM, Johnson CA, Taylor RW, Prokisch H, Doudney K, Mayr JA. Sudden Cardiac Death Due to Deficiency of the Mitochondrial Inorganic Pyrophosphatase PPA2. *Am J Hum Genet.* 2016 Sep 1;99(3):674-682. doi: 10.1016/j.ajhg.2016.06.027. Epub 2016 Aug 11. PMID: 27523597; PMCID: PMC5011043.

Glossary*

Gene	A gene is a unit of heredity that consists of nucleotides or bases. Most genes code for a specific protein leading to a particular characteristic or function. Genes are passed on from parents to offspring.
Variant	A variant is a change or alteration in the genetic sequence at a specific position.
Allele	An allele is one of two DNA sequences occurring at a particular genetic position. An individual possesses two alleles for each gene, one inherited from each parent.
Transcript	A transcript is the single-stranded RNA product synthesized by transcription of DNA. The transcript is then translated to produce protein.
Exon	Exon is the sequence of DNA present in RNA which most often encodes protein.
Intron	Intron is a sequence of DNA between exons that is cut out of the RNA transcript and does not alter the protein sequence. Some intronic sequences are known to affect gene splicing or expression.
Allele Frequency (AF)	Allele frequency is the relative frequency of an allele at a particular locus in the population. Minor allele frequency (MAF) refers to the frequency of the less common allele in a given population.
Heterozygous	Heterozygous genotype occurs when two alleles at a particular genomic position are different. It may include one normal allele and one variant, or two different variants.
Hemizygous	Hemizygous genotype occurs when only one copy of a chromosome is present. This is often used to describe X-linked genes in males since they only have one X chromosome.
Non-Carrier	A non-carrier is an individual that does not have a variant linked to an autosomal recessive disease.
Carrier	A carrier is an individual that has a single copy of variant linked to an autosomal recessive disorder. Carriers do not exhibit symptoms of the condition.
Autosomal Dominant (AD)	AD inheritance refers to a genetic condition that occurs when a variant is present in one copy of a given gene; the individual is heterozygous.
Autosomal Recessive (AR)	AR inheritance refers to a genetic condition that occurs only when variants are present in both copies of a given gene; the individual is homozygous for a variant, or carriers of two different variants on the same gene (compound heterozygous).
X-Linked Dominant (XLD)	XLD inheritance refers to a genetic condition with a mutation in genes on the X chromosome. A single copy of the variant causes the disease in both males and females. Females often present less severe symptoms of the disease.
X-Linked Recessive (XLR)	XLR inheritance references to a genetic condition with a mutation in a gene on the X chromosome. A male carrying a variant will be affected. A female carrying a variant in only one gene is generally unaffected and pass on the disease to their male offspring.

Synonymous Variant	Synonymous variant is a genetic alteration in which a single nucleotide substitution does not lead to an amino acid change. This type of variant is generally benign, unless it affects the splice site.
Missense Variant	Nonsense variant is a genetic alteration that causes the premature termination of a protein, partially or completely inactivating it, resulting in a change or loss of function of the protein. This type of variation is usually pathogenic.
Insertion	Insertion is a genetic alteration that involves the addition or duplication of a segment of DNA (one or more bases).
Deletion	Deletion is a genetic alteration that involves the deletion of a segment of DNA (one or more bases).
Frameshift Variant	Frameshift is an insertion or deletion involving a number of base pairs that is not a multiple of three, consequently disrupting the reading frame. This usually results in the creation of a truncated protein product.
Splice-Disrupt/Intronic Variant	Splice Disrupt and Intronic variants are genetic alterations in the intronic region of the gene. These variants may affect alternative splicing of RNA or alter expression of a gene.
Pathogenic Variant	A pathogenic variant refers to a genetic variant that directly contributes to the development of a disease.
Likely Pathogenic Variant	A likely pathogenic variant refers to a genetic variant that is very likely to contribute to the development of a disease; however, the scientific evidence is currently insufficient to prove this conclusively.
Variant of Uncertain Significance (VUS)	A variant of uncertain significance refers to a genetic variant whose association with disease is not known. There is not enough information to support a more definitive classification of this change.
Likely Benign Variant	A likely benign variant is a genetic variant that is not expected to have a major effect on disease; however, the scientific evidence is currently insufficient to prove this conclusively.
Benign Variant	A benign variant is a genetic variant that does not cause disease.
In Silico Parameter	In silico parameter is the use of computer algorithms that aid in the interpretation of sequence variants. Specifically, it determines impact of the variant on the protein and whether there is an effect on splicing.
Q30%	Q30 refers to the base call accuracy percentage. According to CAP quality measures, Q30% is recommended to be greater than 75%.
Target Coverage at 20x %	Target Coverage at 20x % is the percentage of the target sequence (exome) with coverage greater than 20x. Target coverage at 20x should be greater than 90% as per CAP recommendation.
Mean Region Coverage Depth	Mean Region Coverage Depth indicates how many reads, on average, are likely to be aligned at a given reference base position. Mean Region Coverage Depth cut off should be at 100x.

Rare Exome Variant Ensemble Learner (REVEL)	REVEL is an ensemble method for predicting the pathogenicity of missense variants based on a combination of scores from 13 individual tools: MutPred, FATHMM v2.3, VEST 3.0, PolyPhen-2, SIFT, PROVEAN, MutationAssessor, MutationTaster, LRT, GERP++, SiPhy, phyloP, and phastCons. The REVEL score for an individual missense variant can range from 0 to 1, with higher scores reflecting a greater likelihood that the variant is disease-causing.
CADD	(Combined Annotation Dependent Depletion) score is a tool used to predict the potential impact of genetic variants, particularly single nucleotide polymorphisms (SNPs) and insertions/deletions (INDELs), on the function of genes. The CADD Phred score is based on multiple annotations from various sources, such as conservation scores, sequence features, and functional data. The CADD score typically ranges from 0 to 99, with scores above 20 indicating variants that are more likely to be deleterious (i.e., possibly pathogenic), and scores above 30 suggesting a highly likely damaging effect.
Alpha Missense	refers to a specific category of missense mutations in which a single nucleotide change results in the substitution of one amino acid for another in the protein sequence. The "Alpha" designation typically refers to a subtype or class of missense mutations that are significant due to their effects on protein structure or function. Missense mutations can be classified as either benign or damaging depending on how they affect the protein's folding, stability, and function. The Alpha Missense variant often refers to those that are predicted to have a strong impact on protein function, especially when they occur in highly conserved regions of the protein or affect critical functional domains.

Signature



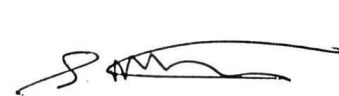
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Appendix 1: Coverage of Genes

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
AARS1	100	AARS2	100	AASS	100
ABAT	100	ABCA1	100	ABCA12	100
ABCA2	100	ABCA3	100	ABCA4	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
ABCA5	100	ABCA7	100	ABCB11	100
ABCB4	100	ABCB6	100	ABCB7	100
ABCC2	100	ABCC6	98.36	ABCC8	100
ABCC9	100	ABCD1	100	ABCD3	100
ABCD4	100	ABCG5	100	ABCG8	100
ABHD16A	100	ABHD5	100	ABL1	100
ACACA	99.89	ACAD8	100	ACAD9	100
ACADM	100	ACADS	100	ACADSB	100
ACADVL	100	ACAT1	100	ACAT2	100
ACBD5	97.87	ACBD6	100	ACO2	100
ACOX1	100	ACOX2	100	ACP5	100
ACSF3	100	ACSL4	100	ACSL5	100
ACTA1	100	ACTA2	100	ACTB	100
ACTC1	100	ACTG1	100	ACTL6B	93.21
ACTN2	100	ACTN4	100	ACVR1	100
ACVR1B	100	ACVRL1	100	ADA	100
ADA2	100	ADAM17	100	ADAM22	100
ADAMTS13	100	ADAMTS2	100	ADAMTS3	100
ADAR	100	ADARB1	100	ADAT3	100
ADCY10	100	ADCY3	100	ADD3	100
ADGRG1	100	ADGRL1	100	ADGRV1	100
ADK	100	ADNP	100	ADORA2A	100
ADPRS	100	ADRA2A	100	ADRA2B	100
ADRB3	100	ADSL	100	ADSS1	100
AEBP1	100	AFF3	100	AFF4	100
AFG3L2	100	AFP	100	AGA	100
AGBL1	100	AGBL5	100	AGGF1	100
AGK	100	AGL	99.66	AGO1	100
AGO2	100	AGPAT2	100	AGPS	99.28
AGR2	100	AGRN	100	AGRP	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
AGTPBP1	100	AGXT	100	AGXT2	100
AHCY	100	AHI1	100	AHR	100
AHSG	100	AICDA	100	AIFM1	100
AIMP1	100	AIMP2	100	AIP	100
AIPL1	100	AIRE	100	AK1	100
AK2	100	AKAP9	100	AKR1D1	99.91
AKT1	100	AKT2	100	AKT3	93.33
ALAD	100	ALAS2	100	ALB	100
ALDH18A1	100	ALDH2	100	ALDH3A2	100
ALDH4A1	100	ALDH5A1	100	ALDH6A1	100
ALDH7A1	100	ALDOA	100	ALDOB	100
ALG1	100	ALG11	100	ALG12	100
ALG13	100	ALG14	100	ALG2	100
ALG3	100	ALG5	100	ALG6	100
ALG8	100	ALG9	100	ALK	100
ALKBH8	100	ALMS1	100	ALOX12B	100
ALOXE3	100	ALPK1	100	ALPK3	100
ALPL	100	ALX4	100	AMACR	100
AMBN	100	AMER1	100	AMFR	100
AMMECR1	100	AMN	100	AMPD1	100
AMPD2	99.84	AMPD3	100	AMT	100
ANAPC1	92.1	ANGPT1	100	ANGPT2	100
ANGPTL3	100	ANGPTL6	100	ANK1	100
ANK2	100	ANK3	100	ANKFY1	100
ANKH	100	ANKLE2	100	ANKRD1	100
ANKRD11	100	ANKRD17	100	ANKRD55	100
ANLN	100	ANO10	100	ANO5	100
ANOS1	100	ANTXR1	100	ANTXR2	100
ANXA11	100	AP1B1	100	AP1G1	100
AP1S1	100	AP1S2	100	AP1S3	87.5

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
AP2M1	100	AP2S1	100	AP3B1	100
AP3B2	100	AP3D1	100	AP4B1	100
AP4E1	100	AP4M1	100	AP4S1	94.27
AP5Z1	100	APC	100	APC2	100
APOA1	100	APOA2	100	APOA5	100
APOB	100	APOC2	100	APOC3	100
APOE	100	APOL1	100	APP	100
APPL1	100	APRT	100	APTX	100
AQP2	100	AQP4	100	AR	100
ARAP3	100	ARCN1	100	ARF1	100
ARFGEF1	100	ARFGEF2	100	ARG1	100
ARHGAP24	100	ARHGAP31	100	ARHGDIA	100
ARHGEF1	100	ARHGEF18	100	ARHGEF9	96.17
ARID1A	100	ARID1B	100	ARID2	100
ARL13B	99.32	ARL2BP	100	ARL3	100
ARL6	100	ARL6IP6	100	ARMC5	100
ARMC9	100	ARNT2	100	ARPC1B	100
ARPC5	100	ARSA	100	ARSB	100
ARSK	100	ARSL	100	ARV1	100
ARVCF	100	ARX	100	ASAH1	100
ASCC3	100	ASCL1	100	ASH1L	100
ASL	100	ASNS	100	ASPA	100
ASPH	100	ASPM	100	ASS1	100
ASXL1	100	ASXL2	100	ASXL3	100
ATAD1	100	ATAD3A	100	ATG7	100
ATIC	100	ATM	100	ATN1	100
ATP10A	100	ATP11A	100	ATP11C	100
ATP13A2	100	ATP1A1	100	ATP1A2	100
ATP1A3	100	ATP2A1	100	ATP2A2	100
ATP2B1	100	ATP5F1A	100	ATP5F1B	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
ATP5F1D	100	ATP5F1E	100	ATP5MK	100
ATP5PO	100	ATP6AP1	100	ATP6AP2	100
ATP6V0A1	100	ATP6V0A2	100	ATP6V0A4	100
ATP6V0C	100	ATP6V1A	100	ATP6V1B1	100
ATP6V1B2	100	ATP6V1E1	100	ATP7A	99.63
ATP7B	100	ATP8A2	100	ATP8B1	100
ATP9A	100	ATPAF2	100	ATR	99.67
ATRX	100	ATXN10	100	ATXN3	96.78
AUH	100	AUTS2	100	AVIL	100
AVP	100	AVPR2	100	AXIN2	100
B2M	100	B3GALNT2	100	B3GALT6	100
B3GAT3	100	B3GLCT	100	B4GALT1	100
B4GALT7	100	B4GAT1	100	B9D1	100
B9D2	100	BAAT	100	BACH2	100
BAG3	99.64	BAG5	100	BANK1	100
BAP1	100	BAZ1B	100	BBIP1	100
BBS1	100	BBS10	100	BBS12	100
BBS2	100	BBS4	100	BBS5	100
BBS7	100	BBS9	100	BCAP31	100
BCAS3	100	BCAT2	100	BCHE	100
BCKDHA	98.81	BCKDHB	100	BCKDK	100
BCL10	100	BCL11B	100	BCL2	100
BCL6	100	BCO1	100	BCOR	100
BCORL1	100	BCR	100	BCS1L	100
BDNF	100	BEST1	100	BGN	100
BICC1	100	BICD2	100	BICRA	100
BIN1	100	BIRC3	98.9	BLK	100
BLM	100	BLNK	100	BLVRA	100
BMP1	100	BMP15	100	BMP2	100
BMP4	100	BMP6	100	BMPER	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
BMPR1A	100	BNC1	100	BOLA3	100
BPGM	100	BPTF	100	BRAF	100
BRAT1	100	BRCA1	100	BRCA2	100
BRD4	99.91	BRIP1	100	BRPF1	100
BSCL2	100	BSND	100	BTD	100
BTK	100	BTNL2	100	BUB1	100
BUB1B	100	BUB3	100	BUD23	100
C12orf57	100	C18orf32	100	C19orf12	100
C1GALT1C1	100	C1QB	100	C1QBP	100
C2orf69	100	C3	100	CA12	100
CA2	100	CA4	100	CA5A	100
CA8	100	CABP4	100	CACNA1A	100
CACNA1B	100	CACNA1C	100	CACNA1D	100
CACNA1E	99.95	CACNA1G	99.91	CACNA1H	100
CACNA1I	100	CACNA1S	100	CACNA2D1	99.79
CACNA2D2	99.34	CACNB4	100	CACNG2	100
CAD	100	CALCRL	100	CALM1	100
CALM2	97.96	CALM3	100	CALR	100
CAMK2A	100	CAMK2B	100	CAMK2G	100
CAMKMT	100	CAMLG	100	CAMSAP1	100
CAMTA1	100	CANT1	100	CAP2	100
CAPN3	100	CAPNS1	100	CAPRIN1	100
CARD10	100	CARD11	100	CARD9	100
CARMIL2	100	CARS1	100	CARS2	100
CARTPT	100	CASK	100	CASP10	100
CASP2	100	CASP8	100	CASQ1	100
CASQ2	100	CASR	100	CASZ1	100
CAT	100	CAV1	100	CAV3	100
CAVIN1	100	CBL	100	CBLB	100
CBLIF	100	CBS	100	CBY1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
CC2D2A	100	CCBE1	100	CCDC134	100
CCDC141	100	CCDC28B	100	CCDC47	100
CCDC78	100	CCDC88A	100	CCDC88C	99.98
CCM2	100	CCN6	100	CCND1	100
CCND2	100	CCNO	100	CCNQ	100
CCR1	100	CCT5	99.05	CD109	100
CD19	100	CD244	100	CD247	100
CD27	100	CD28	100	CD2AP	100
CD320	100	CD3D	100	CD3E	100
CD3G	100	CD40	100	CD40LG	100
CD46	100	CD55	99.22	CD59	100
CD70	100	CD79A	100	CD79B	100
CD81	100	CD96	100	CDAN1	100
CDC40	100	CDC42	100	CDC73	100
CDCA7	100	CDH1	100	CDH11	100
CDH15	100	CDH2	100	CDH23	100
CDHR1	98.04	CDK10	100	CDK13	100
CDK19	100	CDK5	100	CDK6	100
CDK8	100	CDKL5	99.05	CDKN1A	100
CDKN1B	100	CDKN1C	100	CDKN2A	100
CDKN2B	100	CDKN2C	100	CDON	100
CDSN	100	CEACAM3	100	CEACAM6	100
CEBPE	100	CEL	100	CELA2A	100
CELF2	100	CELSR1	100	CENATAC	100
CENPE	100	CEP104	100	CEP120	100
CEP152	100	CEP164	100	CEP19	100
CEP290	100	CEP41	100	CEP57	100
CEP85L	100	CERKL	100	CERS1	100
CERT1	100	CETP	99.85	CFAP418	100
CFB	100	CFH	100	CFHR1	83.33

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
CFHR3	100	CFHR5	100	CFI	100
CFL2	100	CFP	100	CFTR	100
CHAT	100	CHCHD10	100	CHD1	100
CHD2	100	CHD3	99.98	CHD5	100
CHD7	100	CHD8	100	CHEK2	100
CHKB	100	CHMP2B	87.5	CHN1	99.91
CHRNA1	99.97	CHRNA2	100	CHRNA4	100
CHRNA7	91.67	CHRNA2	100	CHRND	100
CHRNG	100	CHST3	100	CHST6	100
CHUK	100	CIC	100	CIDEC	100
CIITA	100	CILK1	100	CISD2	100
CIT	100	CITED2	100	CKAP2L	100
CLCA4	100	CLCF1	100	CLCN2	100
CLCN3	100	CLCN4	100	CLCN5	100
CLCN6	100	CLCN7	100	CLCNKA	100
CLCNKB	100	CLDN10	100	CLDN16	100
CLDN19	100	CLEC7A	100	CLIC2	100
CLIP2	100	CLMP	100	CLN3	96.35
CLN5	100	CLN6	100	CLN8	100
CLP1	100	CLPB	100	CLPP	100
CLPX	100	CLRN1	100	CLTC	100
CLTCL1	100	CLTRN	100	CMPK2	100
CNBP	100	CNGA1	100	CNGB1	100
CNKSR2	100	CNNM2	100	CNOT1	100
CNPY3	100	CNTN1	100	CNTN2	100
CNTNAP1	100	CNTNAP2	100	COA3	100
COA5	100	COA6	100	COA7	100
COA8	100	COG1	100	COG2	100
COG3	100	COG4	100	COG5	100
COG6	100	COG7	100	COG8	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
COL11A1	99.6	COL12A1	100	COL13A1	100
COL18A1	99.99	COL1A1	99.8	COL1A2	99.89
COL25A1	100	COL2A1	100	COL3A1	100
COL4A1	100	COL4A2	100	COL4A3	100
COL4A4	100	COL4A5	100	COL4A6	100
COL5A1	100	COL5A2	100	COL6A1	100
COL6A2	100	COL6A3	100	COL7A1	100
COL8A2	100	COL9A3	100	COLGALT1	100
COLQ	100	COMT	100	COPA	100
COPB1	100	COPB2	100	COQ2	94.08
COQ4	100	COQ5	100	COQ6	100
COQ7	100	COQ8A	100	COQ8B	100
COQ9	100	CORIN	100	CORO1A	88.89
COX10	100	COX11	100	COX14	100
COX15	100	COX16	100	COX20	100
COX4I1	99.83	COX5A	100	COX6A2	100
COX6B1	100	COX7B	100	COX8A	100
CP	100	CPA1	100	CPA6	100
CPAMD8	100	CPE	100	CPLANE1	100
CPLX1	100	CPN1	100	CPOX	100
CPS1	100	CPSF3	100	CPT1A	100
CPT1C	100	CPT2	100	CR2	100
CRADD	100	CRAT	100	CRB1	100
CRB2	100	CRBN	100	CREB3L1	99.97
CREB3L3	100	CREBBP	100	CRELD1	100
CRH	100	CRIPT	100	CRKL	100
CRLF1	100	CRLS1	94.6	CRPPA	100
CRTAP	100	CRX	100	CRYAB	100
CSF1R	100	CSF2RB	100	CSF3R	100
CSGALNAC T1	100	CSNK2A1	100	CSNK2B	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
CSPP1	100	CSRP3	100	CST3	100
CSTB	100	CTBP1	100	CTC1	100
CTCF	100	CTDP1	100	CTH	100
CTLA4	100	CTNNA1	100	CTNNA2	100
CTNNB1	100	CTNNBL1	100	CTNND2	100
CTNS	100	CTPS1	100	CTR9	100
CTRC	100	CTSA	100	CTSD	100
CTSF	100	CTU2	100	CUBN	100
CUL3	99.97	CUL4B	100	CUX2	100
CXCR4	100	CYB561	100	CYB5A	100
CYB5R3	100	CYBA	100	CYBB	100
CYBC1	100	CYC1	100	CYFIP2	100
CYP11A1	100	CYP11B1	100	CYP11B2	100
CYP17A1	100	CYP19A1	100	CYP1B1	100
CYP21A2	90.9	CYP24A1	100	CYP26C1	100
CYP27A1	100	CYP27B1	100	CYP2D6	100
CYP2R1	100	CYP3A4	100	CYP4F22	100
CYP7A1	100	CYP7B1	100	D2HGDH	100
DAAM2	100	DAG1	100	DALRD3	100
DARS1	99.95	DARS2	100	DBH	100
DBR1	100	DBT	99.93	DCAF17	100
DCC	100	DCDC2	100	DCHS1	100
DCLRE1B	100	DCLRE1C	100	DCPS	100
DCTN4	100	DCX	100	DCXR	100
DDB2	100	DDC	99.79	DDOST	100
DDR2	100	DDX3X	100	DDX59	100
DEAF1	100	DEF6	100	DEGS1	100
DENND5A	100	DEPDC5	98.53	DES	100
DGAT1	100	DGCR2	92.31	DGCR6	100
DGCR8	100	DGKE	100	DGUOK	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
DHCR24	100	DHCR7	100	DHDDS	100
DHFR	100	DHPS	100	DHTKD1	100
DHX16	100	DHX30	100	DHX37	100
DHX38	100	DHX9	100	DIAPH1	100
DIO1	100	DIP2B	100	DIS3L2	100
DISP1	100	DKC1	100	DLAT	100
DLD	100	DLG3	100	DLG4	100
DLG5	100	DLK1	100	DLL1	100
DLL4	100	DLST	99.94	DMD	99.83
DMGDH	100	DMP1	100	DMXL2	99.98
DNA2	100	DNAJB11	100	DNAJB4	100
DNAJB6	100	DNAJC19	100	DNAJC21	100
DNAJC3	100	DNAJC30	100	DNAJC5	100
DNAJC6	100	DNASE1	100	DNASE1L3	100
DNASE2	100	DNM1	100	DNM1L	100
DNM2	100	DNMT1	98.34	DNMT3A	100
DNMT3B	99.97	DOCK11	100	DOCK2	100
DOCK6	100	DOCK7	100	DOCK8	100
DOHH	100	DOK7	100	DOLK	100
DPAGT1	100	DPF2	100	DPH1	100
DPH2	100	DPH5	100	DPM1	100
DPM2	100	DPM3	100	DPP9	100
DPYD	99.54	DPYS	100	DPYSL5	100
DRG1	100	DSG1	100	DSG2	100
DSP	100	DSPP	99.67	DST	100
DSTYK	100	DTYMK	99.27	DUOX2	100
DUOXA2	100	DUSP6	100	DUT	100
DVL1	100	DYM	100	DYNC1H1	98.51
DYNC1I2	100	DYNC2H1	100	DYNC2I1	100
DYNC2I2	100	DYRK1A	100	DYRK1B	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
DYSF	100	DZIP1L	100	EARS2	100
EBF3	100	EBP	100	ECE1	100
ECHS1	100	ECM1	100	EDA	100
EDA2R	100	EDAR	100	EDARADD	100
EDEM3	99.88	EDN3	100	EDNRA	100
EED	100	EEF1A2	100	EEF2	100
EFEMP1	100	EFHC1	99.12	EFL1	100
EFTUD2	100	EGF	100	EGFR	100
EGR2	100	EHHADH	100	EHMT1	100
EIF2AK2	99.89	EIF2AK3	100	EIF2AK4	100
EIF2B1	100	EIF2B3	100	EIF2B5	100
EIF2S3	100	EIF3F	100	EIF4A2	100
EIF4H	100	ELAC2	100	ELANE	100
ELF4	100	ELMO2	100	ELN	100
ELOVL4	100	ELP1	100	EMC1	100
EMC10	94.42	EMD	100	EMG1	100
EMILIN1	100	EML1	100	EMP2	100
EN1	100	ENG	100	ENO3	100
ENPP1	100	EOGT	100	EP300	100
EPAS1	100	EPB41	100	EPB42	100
EPCAM	100	EPG5	99.98	EPHB4	100
EPHX2	100	EPM2A	100	EPRS1	100
ERAP1	100	ERBB2	100	ERBB3	100
ERCC1	100	ERCC2	100	ERCC3	100
ERCC4	100	ERCC5	100	ERCC6	100
ERCC8	100	ERG	100	ERGIC1	100
ERLIN2	100	ERMARD	100	ESAM	100
ESCO2	100	ESR1	100	ESS2	100
ETFA	96.67	ETFB	100	ETFDH	100
ETHE1	100	ETS1	100	EXOC2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
EXOC7	100	EXOC8	100	EXOSC3	100
EXOSC5	100	EXOSC8	100	EXOSC9	100
EXT2	100	EXTL3	100	EYS	100
EZH2	100	F10	100	F12	100
F2	100	F5	100	F8	93.93
F9	100	FA2H	99.73	FADD	100
FAH	100	FAM111A	100	FAM111B	100
FAM149B1	100	FAM161A	100	FAM20A	100
FAM20C	100	FAM50A	100	FAN1	100
FANCA	100	FANCB	100	FANCC	100
FANCD2	100	FANCE	100	FANCF	100
FANCG	100	FANCI	100	FANCL	96.47
FANCM	100	FAR1	100	FARS2	100
FARSA	100	FARSB	100	FAS	100
FASLG	100	FASTKD2	100	FAT4	100
FBLN1	100	FBLN5	93.39	FBN1	100
FBP1	100	FBP2	100	FBXL4	100
FBXL7	100	FBXO11	100	FBXO28	100
FBXW7	100	FCGR2A	100	FCGR3A	100
FCGR3B	100	FCSK	100	FDFT1	100
FDX2	100	FDXR	100	FECH	100
FEZF1	100	FGA	100	FGB	100
FGF10	100	FGF12	100	FGF13	100
FGF17	100	FGF20	100	FGF23	100
FGF8	100	FGFR1	100	FGFR2	100
FGFR3	100	FGFRL1	100	FGG	100
FH	100	FHL1	100	FHL2	100
FIBP	100	FIG4	100	FIGLA	100
FILIP1	100	FIP1L1	100	FITM2	100
FKBP10	100	FKBP14	100	FKBP6	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
FKRP	93.22	FKTN	100	FLAD1	100
FLCN	100	FLI1	100	FLII	100
FLNA	100	FLNB	100	FLNC	100
FLRT3	100	FLT1	100	FLT4	100
FLVCR1	100	FLVCR2	100	FMN2	99.98
FMO3	100	FMR1	99.78	FN1	100
FNIP1	100	FOCAD	99.9	FOLR1	100
FOS	100	FOSL2	100	FOXA2	100
FOXC2	100	FOXE1	100	FOXF1	100
FOXH1	100	FOXI1	100	FOXN1	100
FOXP1	100	FOXP2	100	FOXP3	100
FOXRED1	100	FRA10AC1	100	FRAS1	100
FRG1	100	FRMD4A	99.27	FRMD5	100
FRMPD4	100	FRRS1L	100	FSCN2	100
FSHR	100	FTCD	100	FTH1	100
FTL	100	FTO	99.38	FTSJ1	100
FUCA1	100	FUS	100	FUT8	100
FXSD2	100	FZD4	100	FZR1	100
G6PC1	100	G6PD	100	GAA	100
GABBR1	100	GABBR2	100	GABRA1	100
GABRA2	100	GABRA3	100	GABRA5	100
GABRB1	100	GABRB2	100	GABRB3	100
GABRD	97.14	GABRG2	100	GAD1	100
GAL	100	GALC	100	GALE	100
GALK1	100	GALM	100	GALNS	100
GALNT2	100	GALNT3	100	GALT	100
GAMT	100	GANAB	100	GAPVD1	100
GAS1	100	GATA1	100	GATA2	100
GATA3	100	GATA4	100	GATA6	100
GATAD1	100	GATAD2B	100	GATB	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
GATC	100	GATM	100	GBA2	100
GBE1	100	GCDH	100	GCGR	100
GCH1	100	GCK	100	GCLC	100
GCM2	100	GCSH	100	GDF2	100
GDF6	100	GDI1	100	GDNF	100
GEMIN4	100	GET3	100	GET4	100
GFAP	100	GFER	100	GFI1	100
GFM1	100	GFM2	100	GFPT1	100
GFRA1	100	GGCX	100	GGPS1	100
GGT1	100	GHR	100	GHRL	100
GHSR	100	GIPC1	100	GJA1	100
GJA5	100	GJA8	100	GJB3	100
GJB4	100	GJC2	100	GK	100
GLA	100	GLB1	100	GLDC	100
GLE1	100	GLI2	100	GLI3	100
GLIS3	100	GLMN	99.96	GLRA1	100
GLRA2	100	GLRB	100	GLRX5	100
GLS	100	GLUD1	100	GLUL	100
GLYCTK	100	GM2A	100	GMPPA	100
GMPPB	100	GNA11	100	GNA14	100
GNAI1	100	GNAO1	100	GNAQ	100
GNAS	100	GNB1	100	GNB2	100
GNB5	100	GNE	100	GNMT	100
GNPAT	100	GNPNAT1	100	GNPTAB	100
GNPTG	100	GNS	100	GOLGA2	100
GON7	100	GORAB	100	GOSR2	100
GOT2	100	GP1BA	100	GP1BB	100
GP9	100	GPAA1	100	GPC3	100
GPC4	100	GPC6	100	GPD1	100
GPD2	100	GPHN	100	GPI	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
GPIHBP1	100	GPR101	100	GPR161	100
GPR35	100	GPRC5B	100	GPSM2	99.82
GPT2	100	GPX1	100	GRB10	100
GREB1L	100	GREM1	100	GRHL2	100
GRHPR	100	GRIA1	100	GRIA2	100
GRIA3	100	GRIA4	100	GRIK2	100
GRIN1	100	GRIN2A	100	GRIN2B	100
GRIN2D	100	GRIP1	99.97	GRM1	100
GRM7	100	GRN	100	GSN	100
GSR	100	GSS	100	GSTM3	100
GTF2E2	100	GTF2H5	80	GTF2IRD1	100
GTPBP2	100	GTPBP3	100	GUCA1B	100
GUCY1A1	100	GUCY2C	100	GUCY2D	100
GUF1	100	GUSB	100	GYG1	100
GYPE	100	GYS1	100	GYS2	100
GZF1	100	H19	100	H3-3A	100
H4C3	100	H4C5	100	HACD1	100
HACE1	100	HADH	100	HADHA	100
HADHB	100	HAGH	100	HAL	100
HAMP	100	HAND2	100	HAVCR2	100
HAX1	100	HBA1	100	HCCS	100
HCFC1	100	HCN1	100	HCN2	97.36
HCN4	100	HDAC4	100	HDAC8	100
HEATR3	100	HECTD4	100	HECW2	100
HELLS	100	HEPACAM	100	HEPHL1	100
HERC1	100	HERC2	99.9	HESX1	100
HEXA	99.75	HEXB	100	HFE	100
HGD	100	HGSNAT	99.8	HHAT	100
HIBCH	100	HIC1	100	HID1	100
HINT1	100	HIRA	100	HIVEP2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
HJV	100	HK1	99.72	HLA-B	99.89
HLA-DPA1	100	HLA-DPB1	100	HLA-DQA1	100
HLA-DQB1	100	HLA-DRB1	100	HLCS	100
HMBS	100	HMGA1	100	HMGA2	100
HMGCL	100	HMGCR	100	HMGCS2	100
HMOX1	100	HNF1A	100	HNF1B	100
HNF4A	100	HNRNPA1	100	HNRNPA2B1	100
HNRNPC	100	HNRNPDL	99.58	HNRNPH1	100
HNRNPK	100	HNRNPR	100	HNRNPU	100
HOGA1	100	HOXA1	100	HPD	100
HPDL	100	HPGD	100	HPRT1	100
HR	100	HRAS	100	HS3ST6	100
HS6ST1	100	HS6ST2	100	HSD11B1	100
HSD11B2	100	HSD17B10	100	HSD17B3	100
HSD17B4	100	HSD3B2	100	HSD3B7	100
HSPD1	100	HSPG2	100	HTR1A	100
HTRA1	100	HTRA2	100	HTT	100
HUWE1	100	HYAL1	100	HYLS1	100
HYOU1	100	IARS1	100	IARS2	100
IBA57	100	ICOS	100	ICOSLG	100
IDH1	100	IDH2	100	IDH3A	100
IDH3B	100	IDS	100	IDUA	100
IER3IP1	100	IFIH1	100	IFITM5	100
IFNG	100	IFNGR1	100	IFT122	100
IFT140	100	IFT172	100	IFT27	100
IFT43	100	IFT74	99.49	IFT80	100
IFT88	100	IGF1	98.8	IGF1R	100
IGF2	100	IGF2BP2	100	IGFALS	100
IGHG1	100	IGHG2	100	IGKC	100
IGLL1	100	IGSF3	100	IKBKB	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
IKZF1	100	IKZF3	100	IL10	100
IL11RA	100	IL12A	100	IL12B	100
IL12RB1	100	IL17F	100	IL17RA	100
IL17RC	100	IL17RD	100	IL18BP	100
IL1B	100	IL1R1	100	IL1RAPL1	100
IL1RN	100	IL21	100	IL21R	100
IL23R	94.7	IL2RA	100	IL2RB	100
IL2RG	100	IL31RA	100	IL36RN	100
IL37	100	IL6	100	IL6R	100
IL6ST	100	IL7R	100	IMPDH1	100
IMPDH2	100	IMPG1	100	IMPG2	100
INF2	99.46	INPP5E	99.95	INPP5K	100
INPPL1	100	INS	100	INSR	100
INTS1	100	INTS11	100	INTS8	100
INVS	100	IPO8	100	IPW	100
IQCB1	99.67	IQSEC1	99.99	IQSEC2	100
IRAK1	100	IRAK4	100	IREB2	100
IRF1	100	IRF2BP2	100	IRF2BPL	99
IRF3	100	IRF4	100	IRF5	100
IRF6	100	IRF8	100	IRS1	100
IRS2	100	ISCA1	100	ISCA2	100
ISCU	100	ISG15	100	ITCH	100
ITGA2	100	ITGA2B	100	ITGA3	100
ITGA6	100	ITGA7	100	ITGA8	100
ITGAM	100	ITGB2	100	ITGB3	100
ITGB4	100	ITGB6	99.95	ITK	100
ITM2B	100	ITPA	100	ITPR1	100
ITPR2	100	IVD	100	IVNS1ABP	100
IYD	100	JAG1	100	JAG2	99.96
JAK2	100	JAK3	100	JAM2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
JAM3	100	JARID2	100	JAZF1	100
JMJD1C	100	JPH2	100	JRK	99.94
KANK2	100	KANSL1	100	KARS1	100
KAT5	100	KAT6A	100	KAT6B	100
KAT8	100	KATNB1	100	KATNIP	100
KBTBD13	100	KCNA1	100	KCNA2	100
KCNAB2	100	KCNB1	100	KCNC1	100
KCNC2	100	KCNC3	99.72	KCNE1	100
KCNE2	100	KCNE3	100	KCNE5	100
KCNH1	100	KCNH2	100	KCNH5	100
KCNJ1	100	KCNJ10	100	KCNJ11	100
KCNJ13	100	KCNJ16	100	KCNJ2	97.97
KCNJ5	100	KCNJ6	100	KCNK4	100
KCNMA1	100	KCNN2	100	KCNN3	100
KCNN4	100	KCNQ1	99.3	KCNQ2	100
KCNQ3	100	KCNQ5	100	KCNT1	100
KCNT2	100	KCTD1	100	KCTD7	100
KDEL2	100	KDF1	100	KDM1A	99.97
KDM3B	100	KDM4B	100	KDM5A	100
KDM5C	100	KDM6A	100	KDM6B	100
KDSR	100	KHK	100	KIAA0319L	100
KIAA0586	100	KIAA0753	100	KIAA1549	100
KIF11	100	KIF12	100	KIF15	100
KIF1A	100	KIF1B	100	KIF20A	100
KIF22	100	KIF23	100	KIF26A	100
KIF2A	100	KIF4A	100	KIF5A	100
KIF5B	100	KIF5C	100	KIF7	100
KIFBP	98.89	KIRREL1	100	KIT	100
KIZ	100	KL	100	KLF1	100
KLF11	100	KLF13	100	KLF6	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
KLHDC8B	100	KLHL15	100	KLHL3	100
KLHL40	100	KLHL41	100	KLHL7	100
KLHL9	100	KLKB1	100	KLLN	100
KLRC4	100	KMT2A	100	KMT2B	100
KMT2C	98.1	KMT2D	100	KMT2E	100
KMT5B	100	KNG1	100	KNL1	99.35
KNSTRN	100	KPNA3	100	KPTN	100
KRAS	100	KRIT1	100	KRT10	100
KRT13	100	KRT14	100	KRT18	100
KRT2	100	KRT5	100	KRT74	100
KRT9	100	KY	100	KYNU	100
L1CAM	100	L2HGDH	100	LACC1	100
LAGE3	100	LAMA1	100	LAMA2	100
LAMA3	100	LAMA4	100	LAMA5	100
LAMB1	100	LAMB2	100	LAMB3	100
LAMC2	100	LAMC3	99.01	LAMP2	100
LAMTOR2	100	LARGE1	100	LARP7	100
LARS1	100	LARS2	100	LAS1L	98.38
LAT	100	LBR	100	LBX1	100
LCA5	100	LCAT	100	LCK	100
LCP2	100	LCT	100	LDB3	100
LDHA	99.79	LDHB	100	LDHD	100
LDLR	100	LDLRAP1	100	LEMD2	100
LEMD3	100	LEP	100	LEPR	100
LETM1	100	LGI1	100	LGI4	100
LHX1	100	LHX3	100	LHX4	100
LIAS	100	LIFR	100	LIG1	100
LIG3	100	LIG4	100	LIMK1	100
LIMS2	100	LIN28B	100	LINGO1	100
LIPA	99.93	LIPC	100	LIPE	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
LIPN	100	LIPT1	100	LIPT2	100
LMAN1	100	LMAN2L	100	LMBRD1	100
LMBRD2	100	LMF1	100	LMNA	100
LMNB1	100	LMNB2	100	LMO1	100
LMOD2	100	LMOD3	100	LMX1B	100
LNPK	100	LONP1	100	LPIN1	100
LPIN2	100	LPL	100	LRAT	100
LRBA	100	LRP12	100	LRP2	100
LRP5	99.72	LRP6	100	LRPPRC	100
LRRC8A	100	LRRK1	100	LSM11	100
LSS	100	LTBP2	100	LTBP4	100
LTC4S	100	LUZP1	100	LYN	100
LYRM4	98.33	LYRM7	100	LYST	100
LYVE1	100	LYZ	100	LZTFL1	100
LZTR1	100	MACF1	99.99	MAD2L2	100
MADD	100	MAF	98.56	MAFA	100
MAFB	100	MAGED2	100	MAGI2	99.78
MAGT1	100	MAK	100	MALT1	100
MAN1B1	100	MAN2B1	100	MANBA	100
MANF	100	MAP1B	100	MAP2K1	100
MAP2K2	100	MAP3K14	100	MAPK1	100
MAPK10	100	MAPK8IP1	100	MAPK8IP3	100
MAPRE2	100	MAPT	100	MARCHF6	99.55
MARS1	100	MAST1	100	MAST3	100
MAT1A	100	MATR3	100	MAX	100
MB	100	MBD4	100	MBD5	100
MBOAT7	100	MBTPS1	100	MBTPS2	100
MC2R	100	MC4R	100	MCCC1	100
MCCC2	100	MCEE	100	MCFD2	100
MCM10	100	MCM4	100	MCM6	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
MCM9	100	MCOLN1	100	MCPH1	100
MCTP2	100	MCTS1	100	MDFIC	100
MDH1	100	MDH2	100	MECOM	100
MECP2	100	MECR	100	MED11	100
MED12	100	MED12L	100	MED13	100
MED17	100	MED25	100	MED27	100
MEF2A	100	MEF2C	100	MEFV	100
MEGF10	100	MEGF8	100	MEIS2	100
MEN1	100	MERTK	100	MESD	100
MET	100	METTL23	100	METTL27	100
METTL5	100	MFF	100	MFN2	100
MFSD2A	100	MFSD8	100	MGAT2	100
MGME1	100	MGP	100	MIA3	100
MICAL1	100	MICOS13	100	MICU1	100
MID2	100	MIEF2	100	MIF	100
MINPP1	100	MIPEP	100	MIR140	100
MKKS	100	MKRN3	100	MKS1	100
MLC1	100	MLH1	99.82	MLH3	100
MLIP	100	MLX	100	MLXIPL	100
MLYCD	100	MMAA	100	MMAB	100
MMACHC	100	MMADHC	100	MMEL1	100
MMP1	100	MMP14	100	MMP2	100
MMUT	99.87	MN1	100	MOCOS	100
MOCS1	100	MOCS2	100	MOG	100
MOGS	100	MORC2	100	MPC1	100
MPDU1	100	MPDZ	100	MPI	100
MPL	100	MPLKIP	100	MPO	100
MPV17	100	MPZ	100	MRAP	100
MRAS	100	MRE11	100	MRM2	100
MRPL3	100	MRPL39	100	MRPL44	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
MRPS14	100	MRPS16	100	MRPS2	100
MRPS22	100	MRPS23	100	MRPS25	100
MRPS28	83.67	MRPS34	100	MRPS7	100
MRTFA	100	MS4A1	100	MSH2	99.95
MSH3	100	MSH4	100	MSH6	100
MSL3	100	MSMO1	100	MSN	100
MST1	100	MSTO1	89.71	MSX2	100
MT-CO1	100	MT-CO2	100	MT-CO3	100
MT-CYB	100	MTFMT	100	MTHFD1	99.94
MTHFR	100	MTHFS	100	MTM1	100
MTMR14	100	MT-ND1	100	MT-ND2	100
MT-ND3	100	MT-ND5	100	MT-ND6	100
MTNR1B	100	MTO1	99.93	MTOR	100
MTR	100	MT-RNR1	100	MTRR	100
MTSS2	100	MT-TC	100	MT-TE	100
MT-TF	100	MT-TH	100	MT-TK	100
MT-TL1	100	MT-TL2	100	MT-TN	100
MTTP	100	MT-TP	100	MT-TS1	100
MT-TS2	100	MT-TT	100	MT-TV	100
MT-TW	100	MTX2	100	MUC1	98.17
MUC5B	100	MUSK	100	MUTYH	100
MVK	99.39	MYBPC3	100	MYC	100
MYCN	100	MYD88	100	MYF6	100
MYH14	100	MYH3	100	MYH6	100
MYH7	100	MYH9	100	MYL1	100
MYL2	100	MYLK	100	MYMK	100
MYO1E	100	MYO1H	100	MYO5A	100
MYO5B	100	MYO9A	100	MYOC	100
MYOF	100	MYORG	100	MYOT	100
MYPN	100	MYRF	100	MYSM1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
MYT1L	100	MYZAP	100	NAA10	100
NAA15	100	NAA60	100	NAB2	100
NABP1	100	NACC1	100	NADK2	100
NAE1	100	NAF1	100	NAGA	100
NAGLU	100	NAGS	100	NALCN	100
NANS	100	NAPB	100	NARS1	100
NARS2	100	NAT2	100	NAT8L	100
NAXD	100	NAXE	100	NBAS	100
NBEA	100	NBN	100	NCAPD3	100
NCDN	100	NCF2	100	NCF4	100
NCKAP1L	100	NDE1	100	NDN	100
NDNF	100	NDP	100	NDRG1	100
NDST1	100	NDUFA1	100	NDUFA10	100
NDUFA11	100	NDUFA12	100	NDUFA13	100
NDUFA2	100	NDUFA6	100	NDUFA8	100
NDUFA9	100	NDUFAF1	100	NDUFAF2	100
NDUFAF3	100	NDUFAF4	100	NDUFAF5	100
NDUFAF6	100	NDUFAF8	100	NDUFB10	100
NDUFB11	100	NDUFB3	100	NDUFB7	100
NDUFB8	100	NDUFB9	100	NDUFC2	100
NDUFS1	100	NDUFS2	99.51	NDUFS3	100
NDUFS4	100	NDUFS6	100	NDUFS7	100
NDUFS8	100	NDUFV1	100	NDUFV2	100
NEB	90.63	NECAP1	100	NECTIN1	100
NECTIN4	100	NEDD4L	100	NEFH	100
NEFL	100	NEK1	100	NEK10	100
NEK2	96.82	NEK8	100	NEK9	95.83
NELFA	100	NEU1	100	NEUROD1	100
NEUROD2	100	NEUROG1	100	NEUROG3	100
NEXMIF	100	NEXN	100	NF1	99.84

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
NF2	100	NFASC	100	NFE2L2	100
NFIA	100	NFIB	100	NFIX	100
NFKB1	100	NFKB2	100	NFKBIA	100
NFKBIL1	100	NFS1	100	NFU1	100
NGF	100	NGLY1	100	NHEJ1	100
NHLRC1	100	NHLRC2	100	NHP2	100
NIN	100	NIPA1	100	NIPA2	100
NIPAL4	100	NIPBL	99.67	NKX2-1	100
NKX2-5	100	NKX2-6	100	NKX6-2	100
NLGN3	100	NLGN4X	100	NLRC4	100
NLRP1	100	NLRP12	100	NLRP3	100
NMNAT1	100	NNT	100	NOD2	100
NODAL	100	NONO	100	NOP10	100
NOS1	100	NOS1AP	100	NOS3	100
NOTCH1	100	NOTCH2	100	NOTCH3	100
NOVA2	92.52	NPAP1	100	NPAT	100
NPC1	100	NPC2	100	NPHP1	100
NPHP3	100	NPHS1	100	NPHS2	100
NPM1	99.62	NPPA	100	NPR2	100
NPR3	100	NPRL2	100	NPRL3	100
NR0B1	100	NR0B2	100	NR1H4	100
NR2E3	99.96	NR2F1	100	NR3C1	100
NR3C2	99.96	NR4A2	100	NR5A1	100
NRAS	100	NRL	100	NRP1	100
NRP2	100	NRROS	100	NSD1	100
NSD2	99.08	NSDHL	100	NSMCE2	100
NSMCE3	100	NSRP1	97.17	NSUN2	100
NSUN3	100	NSUN6	100	NT5C3A	100
NT5E	100	NTHL1	100	NTNG1	100
NTNG2	100	NTRK1	100	NTRK2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
NUBPL	100	NUMA1	100	NUP107	100
NUP133	100	NUP160	100	NUP205	100
NUP214	100	NUP37	100	NUP85	100
NUP93	99.96	NUS1	100	OAS1	100
OAT	100	OBSCN	100	OCA2	100
OCRL	100	ODC1	100	OFD1	100
OGDH	100	OPA1	100	OPA3	100
OPHN1	100	OPLAH	99.98	ORAI1	99.41
OSGEP	100	OSMR	100	OSTM1	100
OTC	100	OTUD5	100	OTUD6B	100
OTUD7A	100	OTULIN	100	OTX2	100
OVOL2	99.83	OXCT1	100	OXGR1	100
OXR1	100	P3H1	100	P4HA1	100
P4HA2	100	P4HB	100	P4HTM	100
PACS1	100	PACS2	100	PAFAH1B1	100
PAH	100	PAK1	100	PAK2	100
PAK3	100	PALB2	100	PALLD	100
PANK2	100	PANX1	100	PAPPA2	100
PARN	100	PARS2	100	PAX2	100
PAX4	100	PAX6	100	PAX8	100
PBX1	100	PC	100	PCARE	100
PCBD1	90.28	PCCA	100	PCCB	100
PCDH12	100	PCDH19	100	PCDHGC4	100
PCK1	100	PCK2	100	PCLO	100
PCNT	99.99	PCSK1	100	PCSK9	100
PCYT1A	100	PCYT2	100	PDCD1	100
PDCD10	100	PDCD6IP	100	PDE10A	100
PDE2A	100	PDE4D	100	PDE6A	100
PDE6B	100	PDE6D	100	PDE6G	100
PDGFB	100	PDGFRB	100	PDHA1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
PDHB	100	PDHX	100	PDK3	100
PDP1	100	PDPN	100	PDSS1	100
PDSS2	100	PDX1	100	PDZD8	100
PEPD	100	PERCC1	100	PERP	100
PET100	100	PET117	100	PEX1	100
PEX10	100	PEX11B	100	PEX12	100
PEX13	100	PEX14	100	PEX16	100
PEX19	100	PEX2	100	PEX26	100
PEX3	100	PEX5	100	PEX6	100
PEX7	100	PFKM	100	PGAM2	100
PGAP1	100	PGAP2	100	PGAP3	100
PGK1	100	PGM1	100	PGM2L1	100
PGM3	100	PGR	100	PHACTR1	100
PHEX	100	PHF21A	100	PHF6	100
PHGDH	100	PHIP	100	PHKA1	100
PHKA2	100	PHKB	100	PHKG1	100
PHKG2	100	PHLDB1	99.96	PHOX2B	100
PHYH	100	PHYKPL	100	PI4K2A	100
PI4KA	100	PIBF1	100	PIDD1	100
PIEZO1	100	PIEZO2	99.39	PIGA	100
PIGB	100	PIGC	100	PIGF	100
PIGG	100	PIGH	100	PIGK	100
PIGL	100	PIGM	100	PIGN	100
PIGO	100	PIGP	100	PIGQ	99.91
PIGS	100	PIGT	100	PIGU	100
PIGV	100	PIGW	100	PIGY	100
PIK3C2A	100	PIK3CA	100	PIK3CD	100
PIK3CG	100	PIK3R1	100	PIK3R2	100
PIK3R5	100	PITRM1	100	PKD1	100
PKD2	100	PKHD1	100	PKLR	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
PLA2G4A	100	PLA2G6	100	PLA2G7	100
PLAA	100	PLAAT3	100	PLAG1	100
PLAGL1	100	PLAU	100	PLCB1	100
PLCD1	100	PLCE1	100	PLCG1	100
PLCG2	100	PLCH1	100	PLD1	100
PLEC	100	PLEKHG2	100	PLEKHG5	100
PLEKHM1	100	PLG	100	PLIN1	100
PLIN4	100	PLK4	100	PLN	100
PLOD1	100	PLOD2	100	PLOD3	100
PLP1	100	PLPBP	100	PLS3	100
PLVAP	100	PLXNA1	100	PLXND1	100
PMFBP1	100	PML	100	PMM2	100
PMP22	100	PMPCA	100	PMPCB	100
PMS1	94.74	PMS2	100	PNKP	100
PNLIP	100	PNP	100	PNPLA2	100
PNPLA6	99.99	PNPLA8	100	PNPO	100
PNPT1	100	POC1A	100	PODXL	100
POFUT1	100	POGLUT1	100	POGZ	100
POLA1	100	POLD1	100	POLD3	100
POLE	100	POLG	100	POLG2	100
POLR1A	100	POLR2A	100	POLR3A	100
POLR3B	100	POLR3H	100	POLR3K	100
POLRMT	100	POMC	100	POMGNT1	100
POMGNT2	100	POMK	100	POMP	100
POMT1	100	POMT2	100	POPDC3	100
POR	100	POU1F1	100	POU2AF1	100
POU3F3	100	POU3F4	100	POU4F1	97.32
POU6F2	100	PPA2	100	PPARG	100
PPCS	100	PPFIBP1	100	PPIB	100
PPIL1	100	PPM1B	100	PPM1K	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
PPOX	100	PPP1CB	100	PPP1R15B	100
PPP1R17	100	PPP1R21	100	PPP1R3A	100
PPP2CA	100	PPP2R1A	100	PPP2R5D	100
PPP3CA	100	PPT1	100	PQBP1	100
PRCD	100	PRDM13	100	PRDM16	100
PRDM8	100	PRDX1	100	PREPL	100
PRF1	100	PRG4	100	PRICKLE1	100
PRIM1	100	PRKACA	100	PRKAG2	95.61
PRKAR1A	100	PRKAR1B	100	PRKCD	100
PRKCSH	100	PRKCZ	100	PRKDC	100
PRMT7	100	PRNP	100	PROC	100
PRODH	100	PROK2	100	PROKR2	100
PROM1	100	PROP1	100	PRORP	100
PROS1	100	PROX1	100	PRPF3	99.73
PRPF31	100	PRPF4	100	PRPF6	100
PRPF8	100	PRPH2	100	PRPS1	100
PRRT2	100	PRSS1	100	PRSS12	100
PRSS2	100	PRTN3	100	PRUNE1	100
PRX	100	PSAP	100	PSAT1	100
PSEN1	100	PSEN2	100	PSENN	100
PSMB10	100	PSMB4	100	PSMB8	100
PSMB9	100	PSMC3IP	100	PSMD12	100
PSMG2	100	PSPH	100	PSTPIP1	100
PTCD3	100	PTCH1	100	PTEN	99.98
PTF1A	100	PTH	100	PTH1R	100
PTPN1	100	PTPN11	100	PTPN14	100
PTPN2	100	PTPN22	100	PTPN23	100
PTPN3	100	PTPN6	100	PTPRC	100
PTPRO	100	PTRH2	100	PTRHD1	100
PTS	100	PUF60	100	PUM1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
PUS1	100	PUS3	100	PWAR1	100
PWRN1	99.84	PXK	100	PYCR2	100
PYGL	100	PYGM	100	PYROXD1	100
QARS1	100	QDPR	100	QRICH1	100
QRSL1	100	RAB11B	100	RAB18	100
RAB27A	100	RAB34	100	RAB39B	100
RAB3GAP1	96.43	RAB3GAP2	99.59	RABL3	100
RAC1	100	RAC2	100	RAC3	100
RACGAP1	100	RAD21	100	RAD50	100
RAD51	99.39	RAD51C	100	RAF1	100
RAG2	100	RAI1	100	RALA	100
RALGAPA1	100	RANBP2	100	RAP1B	100
RAP1GDS1	100	RAPGEF2	100	RARA	100
RARS1	99.72	RARS2	95.65	RASA1	100
RASA2	100	RASGRP1	100	RB1	100
RBCK1	100	RBL2	100	RBM10	100
RBM20	100	RBM8A	100	RBMX	100
RBP3	100	RBP4	100	RBPJ	100
RD3	100	RDH12	100	RECQL4	100
REEP6	100	REL	100	RELN	100
REN	100	REPS1	100	RERE	100
REST	100	RET	100	RETN	100
RETREG1	100	RFC2	100	RFT1	100
RFWD3	100	RFX5	100	RFX6	100
RFX7	100	RFXANK	100	RFXAP	100
RGR	100	RHAG	100	RHCE	100
RHD	91.67	RHO	100	RHOBTB2	100
RILPL1	100	RIMS2	100	RIN2	100
RINT1	100	RIPK1	100	RIT1	100
RLBP1	100	RMND1	100	RNASEH1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
RNASEH2A	100	RNASEH2B	100	RNASEH2C	100
RNASET2	100	RNF113A	100	RNF125	100
RNF13	100	RNF168	100	RNF2	100
RNF213	100	RNF220	100	RNF31	100
RNF43	100	RNH1	100	RNU7-1	100
ROBO1	100	ROBO3	100	ROGDI	100
ROM1	100	RORA	100	RORB	100
RORC	100	ROS1	100	RP1	100
RP1L1	100	RP2	100	RP9	100
RPA1	100	RPE65	100	RPGR	99.77
RPGRIP1	99.64	RPGRIP1L	96.43	RPIA	100
RPL10	100	RPL11	100	RPL15	100
RPL18	100	RPL26	100	RPL27	100
RPL31	100	RPL35	100	RPL35A	100
RPL3L	100	RPL5	100	RPL8	100
RPL9	100	RPS10	100	RPS14	100
RPS15A	100	RPS19	100	RPS20	100
RPS24	100	RPS26	100	RPS27	100
RPS28	100	RPS29	100	RPS6KA3	100
RPS7	100	RRAGC	100	RRAGD	100
RRAS	100	RRAS2	100	RREB1	100
RRM1	100	RRM2B	100	RSPO1	100
RSRC1	100	RTEL1	100	RTL1	100
RTN2	100	RTN4IP1	100	RTTN	100
RUBCN	100	RUNX1	100	RUSC2	100
RXYLT1	100	RYSR1	100	RYSR2	99.93
RYSR3	100	SAA1	100	SAG	100
SALL4	100	SAMD12	100	SAMD9	100
SAMD9L	100	SAMHD1	100	SAR1B	100
SARDH	100	SARS1	100	SARS2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
SASH3	100	SASS6	100	SAT1	100
SATB1	100	SATB2	100	SBDS	100
SC5D	100	SCAF4	100	SCAPER	100
SCARB2	97.84	SCLT1	100	SCN10A	100
SCN11A	100	SCN1A	100	SCN1B	100
SCN2A	100	SCN3A	100	SCN4A	100
SCN4B	100	SCN5A	100	SCN8A	100
SCN9A	100	SCNN1A	100	SCNN1B	100
SCNN1G	100	SCO1	100	SCO2	100
SCP2	100	SCYL1	100	SCYL2	100
SDC3	98.5	SDCCAG8	100	SDHA	100
SDHAF1	100	SDHAF2	100	SDHB	100
SDHC	100	SDHD	83.33	SDR9C7	100
SEC23B	100	SEC24C	100	SEC24D	100
SEC31A	100	SEC61A1	100	SEC63	100
SECISBP2	100	SELENOI	100	SELENON	100
SEMA3A	100	SEMA4A	100	SEMA4D	100
SEMA6B	100	SEMA7A	100	SEPSECS	100
SERAC1	100	SERPINA1	100	SERPINA6	100
SERPINE1	100	SERPINF1	100	SERPINF2	100
SERPING1	100	SERPINH1	100	SERPINI1	100
SET	100	SETBP1	100	SETD1A	100
SETD1B	100	SETD2	100	SETD5	100
SETX	100	SFTPA2	100	SFTPB	100
SFTPC	100	SFXN4	100	SGCA	100
SGCB	100	SGCD	100	SGCG	100
SGMS2	100	SGPL1	100	SGSH	100
SH2B1	100	SH2B3	100	SH2D1A	100
SH3KBP1	100	SHANK3	99.56	SHARPIN	100
SHH	100	SHOC2	98.42	SHPK	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
SHQ1	100	SHROOM4	100	SI	100
SIK1	100	SIK3	100	SIL1	100
SIM1	100	SIN3A	100	SIN3B	100
SIX3	100	SKI	100	SLC10A1	100
SLC10A2	100	SLC10A7	100	SLC11A1	100
SLC11A2	100	SLC12A1	99.97	SLC12A2	100
SLC12A3	100	SLC12A5	100	SLC12A6	100
SLC13A3	100	SLC13A5	100	SLC16A1	99.87
SLC16A12	100	SLC16A2	100	SLC17A5	100
SLC18A2	100	SLC18A3	100	SLC19A1	100
SLC19A2	100	SLC19A3	100	SLC1A1	100
SLC1A2	100	SLC1A3	100	SLC1A4	100
SLC20A2	100	SLC22A12	100	SLC22A4	100
SLC22A5	100	SLC25A1	100	SLC25A10	100
SLC25A11	100	SLC25A12	100	SLC25A13	100
SLC25A15	100	SLC25A19	100	SLC25A20	100
SLC25A21	100	SLC25A22	100	SLC25A26	100
SLC25A3	100	SLC25A36	100	SLC25A38	100
SLC25A4	100	SLC25A42	100	SLC25A46	100
SLC26A1	100	SLC26A2	100	SLC26A3	100
SLC26A4	100	SLC26A9	100	SLC28A1	99.98
SLC29A3	100	SLC2A1	100	SLC2A2	100
SLC2A3	100	SLC2A9	100	SLC30A10	100
SLC30A2	100	SLC30A8	100	SLC30A9	100
SLC31A1	100	SLC32A1	100	SLC33A1	100
SLC34A1	100	SLC34A2	100	SLC34A3	100
SLC35A1	100	SLC35A2	100	SLC35A3	96.25
SLC35C1	100	SLC35D1	100	SLC36A2	100
SLC37A4	99.97	SLC38A3	100	SLC39A14	100
SLC39A4	100	SLC39A7	100	SLC39A8	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
SLC3A1	100	SLC40A1	100	SLC41A1	100
SLC44A1	100	SLC45A1	100	SLC46A1	99.93
SLC4A1	100	SLC4A10	100	SLC4A11	100
SLC4A2	100	SLC4A4	100	SLC51A	100
SLC51B	100	SLC52A1	100	SLC52A2	100
SLC5A1	100	SLC5A2	100	SLC5A5	100
SLC5A6	100	SLC5A7	100	SLC6A1	100
SLC6A14	100	SLC6A18	100	SLC6A19	100
SLC6A2	100	SLC6A20	100	SLC6A3	100
SLC6A5	100	SLC6A6	99.84	SLC6A8	100
SLC7A14	100	SLC7A6OS	100	SLC7A7	100
SLC7A9	100	SLC9A3	100	SLC9A6	100
SLCO1B1	98.94	SLCO1B3	100	SLCO2A1	100
SLF2	100	SLITRK2	100	SLX4	100
SMAD2	100	SMAD4	100	SMARCA2	100
SMARCA4	100	SMARCA1	100	SMARCB1	100
SMARCC2	100	SMARCD1	100	SMARCE1	100
SMC1A	100	SMC3	100	SMC5	100
SMCHD1	100	SMG9	100	SMO	100
SMPD1	100	SMPD4	100	SMS	100
SNAP25	100	SNAP29	100	SNF8	100
SNIP1	100	SNORD118	100	SNRNP200	100
SNRPN	100	SNTA1	100	SNUPN	100
SNX10	100	SNX14	100	SOD1	100
SON	99.97	SORD	96.44	SORL1	100
SOS1	100	SOS2	99.97	SOST	100
SOX10	100	SOX11	100	SOX18	100
SOX3	100	SOX4	98.44	SOX5	100
SOX9	100	SP110	100	SP7	100
SPARC	100	SPAST	99.54	SPATA7	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
SPEG	100	SPEN	100	SPG11	100
SPG7	100	SPI1	100	SPIB	100
SPIDR	81.82	SPINK1	100	SPINK5	100
SPOP	100	SPP1	100	SPPL2A	100
SPR	100	SPRED1	100	SPRED2	100
SPRY2	100	SPRY4	100	SPTA1	100
SPTAN1	100	SPTB	100	SPTBN1	100
SPTBN4	100	SPTSSA	100	SQOR	100
SQSTM1	100	SRCAP	100	SRD5A2	100
SRD5A3	100	SREBF1	100	SRP19	100
SRP54	100	SRPX2	100	SRSF2	100
SSR4	100	ST3GAL3	100	ST3GAL5	100
STAB1	100	STAC3	100	STAG1	100
STAG2	100	STAMPB	100	STAR	100
STARD7	100	STAT1	96.43	STAT2	100
STAT3	100	STAT4	100	STAT5B	100
STAT6	100	STEAP3	100	STEEP1	100
STIL	100	STIM1	100	STING1	100
STK11	100	STK4	100	STOX1	92.8
STRADA	100	STS	100	STT3A	100
STT3B	100	STUB1	99.95	STX11	100
STX16	100	STX1A	100	STX1B	100
STX3	100	STX5	100	STXBP1	100
STXBP2	100	SUCLA2	99.77	SUCLG1	100
SUFU	100	SUGCT	98.59	SULT2B1	100
SUMF1	100	SUN5	100	SUOX	100
SUPT16H	100	SURF1	100	SV2A	100
SVBP	100	SVEP1	100	SVIL	100
SYK	100	SYN1	100	SYNE1	100
SYNE2	100	SYNGAP1	100	SYNJ1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
SYP	100	SYT2	100	SZT2	98.44
TACO1	100	TACR3	100	TACSTD2	100
TAF1	100	TAF1A	100	TAF4	98.51
TAFAZZIN	100	TALDO1	100	TAMM41	100
TANC2	100	TANGO2	100	TAOK1	100
TAPT1	100	TARDBP	100	TARS1	100
TARS2	100	TASP1	100	TAT	100
TBC1D20	100	TBC1D23	100	TBC1D24	100
TBC1D2B	100	TBC1D8B	100	TBCD	100
TBCE	100	TBCK	100	TBK1	100
TBL1X	100	TBL1XR1	100	TBL2	100
TBP	100	TBR1	100	TBX1	99.95
TBX19	100	TBX20	100	TBX3	100
TBX5	100	TCAP	100	TCEAL1	100
TCF20	100	TCF3	100	TCF4	100
TCF7L2	100	TCIRG1	100	TCN2	100
TCTN1	100	TCTN2	100	TCTN3	100
TDO2	100	TDP1	100	TDP2	100
TECPR2	100	TEFM	100	TEK	100
TELO2	100	TENT5A	100	TERC	100
TERT	100	TET2	100	TET3	100
TF	100	TFAM	100	TFAP2A	100
TFE3	100	TFG	100	TFR2	100
TFRC	100	TG	100	TGDS	100
TGFB1	100	TGFB1	100	TGFBR2	100
TGFBR3	100	TGIF1	100	TGM1	100
TH	100	THBD	100	THOC2	100
THPO	100	THRB	100	THSD1	100
THUMPD1	100	TIA1	100	TIAM1	100
TICAM1	100	TIE1	100	TIMM22	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
TIMM50	100	TIMM8A	100	TIMMDC1	100
TINF2	100	TJP2	100	TK2	92.86
TKFC	100	TKT	100	TLK2	100
TLL1	100	TLR3	100	TLR4	100
TLR7	100	TLR8	100	TMC6	100
TMEM106B	100	TMEM126B	100	TMEM127	100
TMEM138	100	TMEM147	100	TMEM163	100
TMEM165	100	TMEM216	100	TMEM218	100
TMEM222	100	TMEM231	100	TMEM237	100
TMEM260	100	TMEM270	100	TMEM38B	100
TMEM43	100	TMEM67	100	TMEM70	100
TMEM94	100	TMLHE	100	TMPO	100
TMPRSS15	99.97	TMPRSS6	99.96	TMTC3	99.7
TMX2	100	TNFAIP3	100	TNFRSF11A	100
TNFRSF11B	100	TNFRSF13B	100	TNFRSF13C	100
TNFRSF1A	100	TNFRSF1B	100	TNFRSF9	100
TNFSF11	100	TNFSF15	100	TNFSF4	100
TNIP1	100	TNNC1	100	TNNI3	100
TNNT1	97.68	TNNT2	99.72	TNPO2	100
TNPO3	99.52	TNRC6B	100	TOE1	100
TOGARAM1	100	TOM1	100	TOMM40	100
TOMM7	100	TONSL	100	TOP3A	100
TOPORS	100	TOR1AIP1	100	TP53	100
TP53RK	100	TP73	100	TPK1	100
TPM1	100	TPM2	100	TPM3	100
TPMT	100	TPO	100	TPP1	100
TPP2	100	TPRKB	100	TRAC	99.98
TRAF3	100	TRAF3IP2	100	TRAF6	100
TRAF7	100	TRAK1	100	TRANK1	100
TRAPPC10	100	TRAPPC11	100	TRAPPC12	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
TRAPPC2	100	TRAPPC2L	100	TRAPPC4	100
TRAPPC6B	100	TRAPPC9	100	TRDN	100
TREH	100	TREM2	100	TRHR	100
TRIM28	100	TRIM32	100	TRIM37	100
TRIM71	100	TRIM8	100	TRIO	99.99
TRIP11	100	TRIP12	100	TRIP13	100
TRIP4	100	TRIT1	100	TRMT1	100
TRMT10A	100	TRMT10C	100	TRMT5	100
TRMU	100	TRNT1	100	TRPC6	100
TRPM3	100	TRPM6	100	TRPS1	100
TRPV3	100	TRPV4	100	TRPV6	100
TRRAP	100	TRU-TCA1-1	100	TSC1	99.96
TSC2	100	TSEN15	100	TSEN2	100
TSEN34	100	TSEN54	100	TSFM	100
TSHB	100	TSHR	100	TSPAN12	100
TSPOAP1	100	TSPYL1	100	TSR2	100
TTC19	100	TTC5	100	TTC7A	100
TTC8	100	TTI1	100	TTN	99.33
TTPA	99.91	TTR	100	TUB	100
TUBA1A	100	TUBA8	100	TUBB	100
TUBB2A	100	TUBB2B	100	TUBB3	100
TUBB4A	100	TUBB4B	100	TUBG1	100
TUBGCP4	100	TUBGCP6	100	TUFM	100
TULP1	100	TULP3	100	TWNK	100
TXN2	100	TXNDC15	100	TXNRD2	100
TYK2	100	TYMP	100	TYMS	100
TYROBP	100	U2AF2	100	UBA1	100
UBA5	100	UBAC2	100	UBAP1	100
UBAP2L	100	UBE2A	100	UBE2L3	100
UBE2T	100	UBE3A	100	UBE3B	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
UBE4A	100	UBE4B	100	UBR1	100
UBR7	100	UBTF	100	UCP2	100
UCP3	100	UFC1	100	UFD1	100
UFM1	100	UFSP2	100	UGDH	100
UGP2	100	UGT1A1	100	UHRF1	100
UMOD	100	UMPS	100	UNC13D	100
UNC45A	100	UNC80	100	UNC93B1	100
UNG	100	UPB1	100	UPF3B	100
UQCC2	100	UQCRB	100	UQCRC2	100
UQCRFS1	100	UQCRH	100	UQCRQ	99.83
UROC1	100	UROD	100	UROS	100
USB1	100	USF3	100	USH2A	100
USP18	90	USP45	100	USP48	100
USP53	100	USP7	100	USP8	100
USP9X	100	UVSSA	100	VAC14	100
VAMP1	100	VAMP2	100	VAPB	100
VARS1	100	VARS2	100	VCL	100
VCP	100	VDR	100	VEGFC	99.65
VEZF1	100	VHL	100	VIPAS39	100
VLDLR	100	VMA21	100	VPS11	100
VPS13A	100	VPS13B	100	VPS13D	100
VPS16	100	VPS33A	99.74	VPS33B	100
VPS35L	100	VPS37D	100	VPS41	100
VPS45	100	VPS4A	100	VPS50	100
VPS51	100	VPS53	99.95	VRK1	100
VSX1	100	WAC	100	WARS1	96.04
WARS2	100	WAS	100	WASF1	100
WBP4	100	WDPCP	100	WDR1	100
WDR11	100	WDR19	99.9	WDR26	100
WDR35	100	WDR37	100	WDR4	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
WDR45	100	WDR45B	100	WDR62	100
WDR73	100	WDR81	100	WFS1	100
WIPF1	100	WLS	100	WNK1	100
WNK4	100	WNT1	100	WNT11	100
WNT7A	100	WNT9B	100	WRAP53	100
WRN	100	WT1	100	WWOX	100
XDH	100	XIAP	100	XK	100
XPA	100	XPC	100	XPNPEP2	100
XPNPEP3	100	XPR1	100	XRCC2	100
XRCC4	100	XYLT1	100	XYLT2	100
YAP1	100	YARS1	100	YARS2	100
YEATS2	100	YIF1B	100	YIPF5	100
YME1L1	100	YRDC	99.31	YWHAE	100
YWHAG	100	YY1	100	ZAP70	100
ZBTB16	100	ZBTB18	99.97	ZBTB20	100
ZBTB24	100	ZBTB7A	100	ZC4H2	100
ZDHHC9	100	ZEB1	100	ZEB2	100
ZFHX2	100	ZFP57	95.45	ZFX	100
ZFYVE19	100	ZFYVE26	100	ZIC2	100
ZMIZ1	100	ZMPSTE24	100	ZMYM2	100
ZMYM3	100	ZMYND11	100	ZNF142	100
ZNF292	100	ZNF341	100	ZNF408	100
ZNF423	100	ZNF513	100	ZNF592	100
ZNF668	100	ZNF687	100	ZNF699	100
ZNFX1	100	ZNHIT3	100	ZNRF3	99.37
ZPR1	100	ZSWIM6	99.29	ZSWIM7	100

Name: **B/O Sasi Priya**
Age: **3 days**
Sex: **Female**
Referring Clinician: **Dr. S Kiruthika**
Test Required: **Mitochondrial Genome Sequencing**
Location: **Salem Genetics Center,Salem**

Case ID: **50800110018**
Sample Type: **Peripheral Venous Blood**
Sample Collection Date: **22 Aug 2025**
Sample Receipt Date: **23 Aug 2025**
Report Date: **13 Sep 2025**
Version: **I**

CLINICAL INFORMATION/HISTORY

SASI PRIYA is non-consanguineously married to MUNI PRASAD. The couple's first pregnancy was a missed abortion. G2- female baby died at 3 months (sudden death) and was suspected to have increased startle response, seizures, and cardiomegaly. G3- female baby, intra uterine death at 32 weeks of gestation, maternal history of chicken pox at 23-24 weeks of gestation. G4- missed abortion. G5- female baby born through normal vaginal delivery, no history of NICU stay. Neurosonogram of the baby was Normal, USG abdomen- normal. Normal limbs, suspected prominence of clitoris. Couple karyotyping report normal. The baby (G5) is being evaluated for gene variations relating to the reported phenotype.

The following HPO terms were used for analysis: inborn errors of metabolism, sudden death, neonatal death, Exaggerated startle response, seizures, cardiomegaly, intra uterine death, abnormal size of the clitoris, abnormal clitoris morphology, Abnormal size of the clitoris

RESULT SUMMARY

No Pathogenic or Likely Pathogenic variants causative of the reported phenotype were identified

*Correlation with clinical profile and family history is required

FINDINGS RELATED TO PHENOTYPE

No variations likely to be causative of the reported phenotypes/clinical symptoms were detected.

The mitochondrial genome was completely covered (genes listed below)

(MT-TF, MT-RNR1, MT-TV, MT-RNR2, MT-TL1, MT-ND1, MT-TI, MT-TQ, MT-TM, MT-ND2, MT-TW, MT-TA, MT-TN, MT-TC, MT-TY, MT-CO1, MT-TS1, MT-TD, MT-CO2, MT-TK, MT-ATP8, MT-ATP6, MT-CO3, MT-TG, MT-ND3, MT-TR, MT-ND4L, MT-ND4, MT-TH, MT-TS2, MT-TL2, MT-ND5, MT-ND6, MT-TE, MT-CYB, MT-TT and MT-TP genes).

RECOMMENDATIONS

The interpretation of this result should be done in the context of this individual's clinical and biochemical profile. Genetic counseling is recommended.

NOTES:

A list of variants identified in this individual is available upon request. Variants are evaluated by their reported frequency in databases such as mitomap. Variants that have a population frequency greater than expected given the prevalence of the disease in the general population are considered to be benign variants. Silent variants (Synonymous variants) are not reported unless known to be pathogenic or other evidence suggests potential disruption of splicing. The correlation of the genotype to the phenotype is based on our current understanding of the genes involved. Classification of the variants will be done using ACMG guidelines (1). These interpretations may change over time as more information about the gene(s) and this individual's clinical phenotype becomes available. Raw sequencing data is available upon request.

Variant Classification based on ACMG recommendations:

Term	Definition
Pathogenic Variant	A disease causing variation in a gene which can explain the patient's symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed.
Likely Pathogenic Variant	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance (VUS)	A variant has been detected, but it is difficult to classify it as either pathogenic (disease-causing) or benign (non-disease-causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. Their significance can probably be assessed only with time, subject to the availability of scientific evidence.

METHODOLOGY

Mitochondrial genome sequencing encompasses careful and selective amplification and sequencing of the complete mitochondrial genome. There are 37 genes in the mitochondrial genome that are essential for normal mitochondrial function. The entire mitochondrial genome is amplified in two fragments using specific primers by long-range PCR followed by library preparation using the Roche Kapa Hyperplus kit. The libraries were sequenced to mean 1000-2000X coverage on the Illumina sequencing platform. The paired-end reads were aligned to the reference human genome Feb 2009 release downloaded from the UCSC database (GRCh37/hg19)[2,3,4]. We have used the Revised Cambridge reference sequence (rCRS) as the mitochondrial reference genome (<http://www.mitomap.org/MITOMAP/Human/Mitoseq>). We follow the GATK best practices framework for the identification of variants. Clinically relevant mutations were annotated using published variants in literature and the MitoMap database [5]. Only non-synonymous variants found in the sample were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

LIMITATIONS

Test results are interpreted in the context of clinical findings, family history and other biochemical data. This assay cannot detect variants in regions of the genome that are not amenable to NGS, such as areas containing large numbers of tandem repeats.

DISCLAIMER

- Mutations have not been confirmed by Sanger sequencing.
- Insufficient clinical details may lead to missing of causative pathogenic variants from reporting.

REFERENCES

1. Richards S. et al., Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology, Genet Med., 17(5):405-24, 2015.
2. Li, H. and R. Durbin. Fast and accurate long-read alignment with Burrows-Wheeler transform. Bioinformatics, 2010. 26(5): p. 589-95.
3. Meyer, L.R., et al., The UCSC Genome Browser database: extensions and updates 2013. Nucleic Acids Res, 2013. 41(D1): p. D64-9.
4. McKenna, A., et al., The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. Genome Res, 2010. 20(9): 1297-303.
5. <http://www.mitomap.org/MITOMAP>

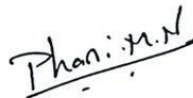
Signature



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